

Aircraft Design I – MMAE 414

Fall Semester 2025 version 1.1

Week	Date Monday	Monday – lecture 10:00 – 11:15	Wednesday – “Lab” 10:00 – 11:15	Reading & Homework Assignments
1	8/18	Chpt. 2 – Intro & expectations - Aircraft Design Milestones	Lab 1 - Xflr5.com YouTube videos 1 - 4	Read Chpt. 1 & 2 – Design Read Chpt. 3 Sizing
2	8/25	Chpt. 3 – Phases of design and weight estimate	Lab 2 – YouTube video 6 - 9	Read Chpt. 4 Airfoil, wing, tail
3	9/1 (labor day)	Labor day (no class) Friday - lecture	Lab 3 - YouTube video 11 - 15 – stability, static and dynamic – Elevator, aileron, rudder additions to xflr5	Read Chpt. 5 Wing loading
4	9/8	Chpt. 4 – Airfoil, Wing Tail geometry Chpt. 5 - T/W, P/W and W/S	Lab 4 Matlab/Simulink Flight Sim 3DOF with elevator -	Read Chpt. 6 Initial sizing
5	9/15	Chpt. 6 - Initial sizing	Quiz #1	Read Chpt. 9 & 10 Passengers, payload and fuel
6	9/22	Chpt. 9 & 10 - Crew, Passengers, Payload, & Propulsion and fuel	Lab 5 Matlab/Simulink Flight Sim- 3DOF Design Progress Report	Read Chpt. 11 Landing gear and Chpt. 14 Structures
7	9/29	Chpt. 11 Landing gear Chpt. 19 Trade studies – and Chpt. 18 Cost Analysis	Lab 6 - Matlab/Simulink Flight Sim – rudder & aileron control 6DOF	Read Chpt. 19 and Chpt. 18.
8	10/6	Midterm – Conceptual Design presentations 1	Midterm – Conceptual Design presentations 2	Read Chpt. 16 Stability control
9	10/13 Fall break	Monday – no class Wed – Stability & control	Lab 7 - Matlab/Simulink Flight Sim – 3DOF feedback trajectory control	
10	10/20	NATO Quiz #2	NATO Lab 8 - Matlab/Simulink Flight Sim –	Read Chpt. 17 Performance
11	10/27	Chpt. 17 Feedback control	Lab 9 – Longitudinal flight control with Simulink - 6DOF feedback trajectory control	Read journal article on MDAO
12	11/3	Multi-disciplinary design analysis and optimization MDAO	Lab 10 – Roll angle feedback control – fly airplane in a circle	

13	11/10		Risk analysis	
14	11/17	Risk analysis	Quiz #3	
15	11/24		Thanksgiving	
16	12/1	Final Presentations - 1	Final Presentations - 2	
17	12/8	Finals Week		

Text book: "Aircraft Design: A Conceptual Approach," 6th edition, by Daniel Raymer, AIAA Education Series, 2018, Isbn 978-1-62410-490-9

References:

www.xflr5.com has the xflr5 code and tutorials required for this course.

An intro to aircraft design using Matlab/Simulink can be found here.

<http://www.mathworks.com/company/newsletters/articles/accelerating-flight-vehicle-design.html>

Prerequisites: MMAE 302 OR 304, MMAE 312, MMAE 352

Corequisites: MMAE 410

Lab: All students must bring a laptop computer with the XFLR5 software installed.

Instructional YouTube videos can be found on the same website.

Grading	Instructor
Midterm Presentation (0.3), Report (0.55), Evals (0.15)	20% 
3 Quizzes	35% Room 209-C REC bldg.
Final Presentation (0.3), Report (0.55), Evals (0.15)	25% Office hours TR 9:30 - 11:30
Homework	15%
Participation	5% (based on attendance and peer evals)

The midterm grade and the final project grades are based on the written report (55%), the presentation (30%), and peer evaluations within each group (15%).

Score > 90 = A
80 > Score > 89 = B
70 < Score > 79 = C
60 < Score < 69 = D
Score < 60 = E

Course Objectives and Student Outcomes

Course Outcomes:

Upon the course completion, students will be able to:

- Develop an ability to integrate the knowledge and methods from previous courses on the fundamental engineering science pertaining to the design of aircraft
- Understand the basic elements of the aircraft design process, consisting of: (a) requirement definition; (b) initial sizing; (c) configuration layout; (d) trade study analysis; (e) application of FAR standards (Title 14 - CFR) to the design
- Recognize the limitations of available methods for the analysis of aircraft designs
- Use available methods for the prediction of aircraft performance to obtain aerodynamic stability and control
- Understand the iterative nature of aircraft design
- Evaluate aerodynamic performance of the design using a flight simulation
- Emphasize the importance of written and graphical communication

Relationship of Course to ABET Student Outcomes:

ABET SO #	Student Outcome
1	An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2	An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3	An ability to communicate effectively with a range of audiences
5	An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
7	An ability to acquire and apply new knowledge as needed, using appropriate learning strategies

ADA statement

Reasonable accommodations will be made for students with documented disabilities. In order to receive accommodations, students must obtain a letter of accommodation from the Center for Disability Resources. The Center for Disability Resources (CDR) is located in Life Sciences

Code of Academic Honesty (see attachment for MMAE policy)

IIT expects students to maintain high standards of academic integrity. Students preparing for the practice of a profession are expected to conform to a code of integrity and ethical standards commensurate with the high expectations society places on practitioners of a learned profession. No student may seek to gain an unfair advantage over another. The Code of Academic Honesty is explained in the IIT Student Handbook and all students are expected to know and adhere to this code.

Collaboration Policy: Collaboration on homework is permitted, but you must produce your own unique solutions. Any use of homework or quiz solutions from prior years is prohibited. Collaboration of any kind on quiz is prohibited.

Classroom Expectations Regarding Diversity, Equity, and Inclusion:

The MMAE Department statement on Diversity, Equity, and Inclusion including the incident reporting process is here: (<https://www.iit.edu/sites/default/files/2021-01/MMAE-Statement-On-Diversity-Equity-and-Inclusion.pdf>). I expect everyone to treat each other with empathy and civility both in person and online.

Academic Integrity and Collaboration Policy:

Departmental policies on academic honesty are consistent with those of the University. The Department policies are provided here: (https://www.iit.edu/sites/default/files/2021-01/honesty_policy_-_mmae.pdf).

Additional learning resources available at IIT are here: (<https://www.iit.edu/mmae/student-resources>).

**DEPARTMENT OF MECHANICAL, MATERIALS, AND
AEROSPACE ENGINEERING**

POLICY ON ACADEMIC HONESTY

The Department of Mechanical, Materials, and Aerospace Engineering at Illinois Institute of Technology considers academic training to be apprenticeship for practice in the professions. Students are expected to demonstrate a code of moral integrity and ethical standards commensurate with the high expectations that society places upon professional practice.

Accordingly, it is the policy of the department to maintain the highest standard of academic honesty and integrity. The department prohibits any form of cheating that contributes, or is intended to contribute, to a student's grade in a course. This includes copying or unauthorized collaboration on homework, reports, computer programs, projects, etc., collaboration of any kind on an examination or quiz, or the unauthorized possession or use of any material or machine-stored information during an examination. It is a violation for a student, whether or not currently enrolled in the university, to knowingly engage or attempt to engage in the acquisition, without permission, of tests, answer sheets, problem solutions or other academic material when such material has been withheld from distribution by the instructor.

Students judged guilty of such offenses are subject to either of the following sanctions:

1. **Expulsion from a course.** The student is assigned a punitive failing grade of E for the course and can no longer participate in the course or receive evaluation of coursework from the instructor.
2. **Reduction in Grade.** A reduction in grade for the assignment or exam involved or for the course may be applied.

In addition, the incident will be reported to the Designated Dean for Academic Discipline, who may make a recommendation for further disciplinary action. The report of the incident will be placed in the student's permanent academic record. An offense may warrant a recommendation to initiate dismissal proceedings.

This policy will apply not only to a student guilty of any form of academic dishonesty, but also to any party who knowingly allows his or her work to be plagiarized.

This policy is consistent with the IIT Code of Academic Honesty
(http://www.iit.edu/student_affairs/handbook/)